

GOOGLE CYBERSECURITY CERTIFICATE

Threat Modeling — PASTA

Google Cybersecurity Professional Certificate · Study Guide

A full 7-stage PASTA (Process for Attack Simulation and Threat Analysis) threat model for a sneaker company's mobile app — from business objectives through API-focused threat and vulnerability analysis to recommended controls.

Track	Google Cybersecurity Professional Certificate
Course	Assets, Threats, and Vulnerabilities
Framework	PASTA — 7 stages
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1 Stages I–IV



PASTA — 7 Stages

1. I. BUSINESS & SECURITY OBJECTIVES

Member profiles, financial transactions, PCI-DSS compliance

2. II. TECHNICAL SCOPE

API, PKI, SHA-256, SQL – API prioritized

3. III. APPLICATION DECOMPOSITION

Data flow diagram: users, systems, APIs

4. IV. THREAT ANALYSIS

Injection, session hijacking

Stage I — Business & Security Objectives

Objective

Users can create member profiles internally or by connecting external accounts

The app must process financial transactions

The app must be compliant with PCI-DSS

Stage II — Technical Scope

Technology	Notes
API	Facilitates data exchange between customers, partners, employees — handles sensitive data across a large attack surface; prioritized for analysis
PKI	Manages encryption and digital certificates
SHA-256	Cryptographic hashing
SQL	Database querying and management

NOTE APIs were prioritized because they connect multiple systems and users, handle sensitive data at scale, and expose a larger attack surface than the other technologies in scope.

Stage IV — Threat Analysis

Threat type	Description
Injection	Malicious input (e.g. SQL injection) exploiting unsanitized queries
Session hijacking	Attacker intercepts a valid session token to impersonate a legitimate user

2 Stages V–VII

Stage V — Vulnerability Analysis

Vulnerability	Description
Lack of prepared statements	SQL queries not using parameterized inputs, leaving them open to injection
Broken API token	Improperly validated or expired API tokens that can be exploited

Stage VI — Attack Modeling

An attack tree was built mapping the concrete exploitation paths available for the identified threats — from an unauthenticated request, through the vulnerable API endpoints, to data exposure or session takeover.

Stage VII — Risk Analysis & Security Controls

Control	Purpose
SHA-256	Securely hash sensitive data like passwords
Incident response procedures	Defined playbooks for responding to breaches
Password policy	Enforce strong, unique passwords across accounts
Principle of Least Privilege	Limit user and service access to only what is required

3 Quick Reference



PASTA framework — quick reference

PASTA — 7 stages:

- I. Define business and security objectives
- II. Define technical scope
- III. Decompose the application
- IV. Threat analysis
- V. Vulnerability analysis
- VI. Attack modeling
- VII. Risk analysis & security controls